Sampling Methods & Bias

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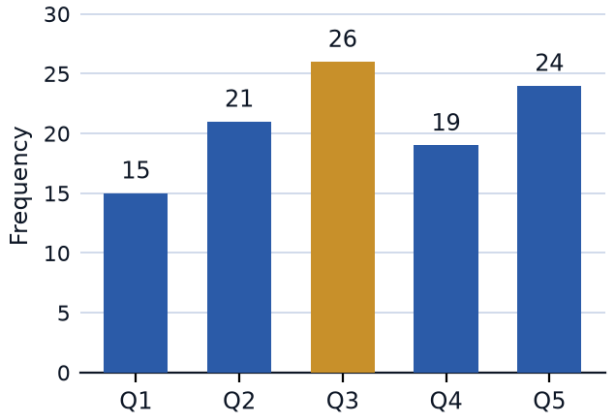
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Identifying sampling methods

A school assigns each student a number from 1 to 800, then uses a random number generator to select 40 numbers, surveying those students. Identify this sampling method.

2 [Foundational] • Target: Identifying sampling methods

A researcher divides a school into grade levels (9th, 10th, 11th, 12th), then takes a simple random sample of 20 students from each grade. Identify this sampling method and explain why it might be preferred over an SRS of the whole school.



3 [Foundational] • Target: Cluster sampling and systematic sampling

A researcher randomly selects 5 entire homeroom classes out of 40 in a school, and surveys every student in those 5 classes. Identify this sampling method, and contrast it with stratified sampling.

4 [Foundational] • Target: Cluster sampling and systematic sampling

A quality-control inspector checks every 25th item coming off a production line. Identify this sampling method.

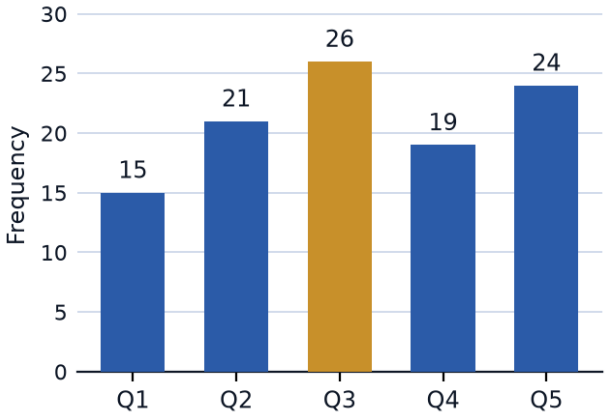
5 [Foundational] • Target: Sources of bias

A survey about a new school policy is voluntarily filled out online, and only 12% of students respond. Identify the type of bias most likely present, and explain why the results may not represent the whole student body.

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6 [Foundational] • Target: Sources of bias

A phone survey about internet usage habits is conducted using only landline phone numbers. Identify the type of bias this introduces.



7 [Foundational] • Target: Wording and response bias

A survey question reads: "Don't you agree that the wasteful new cafeteria policy should be reversed?" Explain why this question is problematic, and rewrite it in a neutral form.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Wording and response bias

Explain why respondents might not answer a sensitive question (such as about illegal activity) truthfully in a face-to-face interview, and name this type of bias.

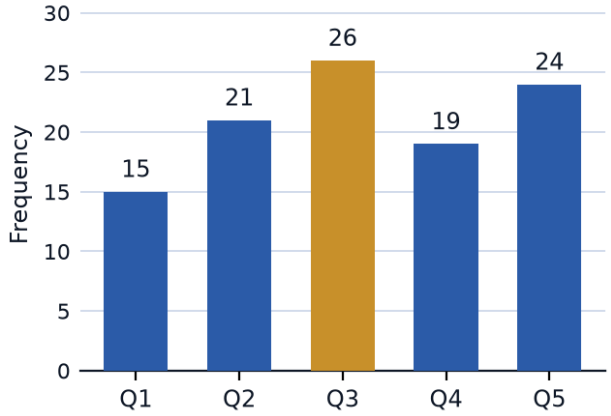
9 [Intermediate] • Target: Comparing sample size and bias

Explain why increasing the sample size does not fix a biased sampling method, using the voluntary response survey from earlier as an example.

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10 **[Intermediate]** • **Target: Designing a sampling plan**

A city has 4 distinct neighborhoods with very different income levels, and the mayor wants to survey residents about a new tax proposal in a way that fairly represents each neighborhood. Recommend a sampling method, and describe how it would be carried out.



11 **[Intermediate]** • **Target: Convenience sampling and its pitfalls**

A researcher stands outside a gym and surveys the first 50 people who walk out about their exercise habits, intending to represent the whole city's exercise habits. Identify this sampling method and explain the flaw in generalizing its results to the whole city.

12 **[Intermediate]** • **Target: Comparing sampling methods side by side**

A national retailer wants to estimate average customer satisfaction across all 500 of its stores. Compare how a cluster sample (randomly selecting 20 whole stores and surveying all customers there) and a stratified sample (grouping stores by region, then sampling customers from every region) would each be carried out, and state one advantage of each approach.

13 **[Intermediate]** • **Target: Identifying the population and sample**

A university wants to know the average commute time for all 30,000 enrolled students, and surveys a random sample of 400 students. Identify the population and the sample in this scenario.

14 **[Intermediate]** • **Target: Identifying the population and sample**

A national news outlet runs a text-in poll during a live broadcast, asking viewers to text "YES" or "NO" on a controversial policy question, and reports the results as representing "public opinion." Identify every source of bias you can find in this scenario, and explain why the reported percentages should not be trusted as an accurate reflection of the general public's views.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

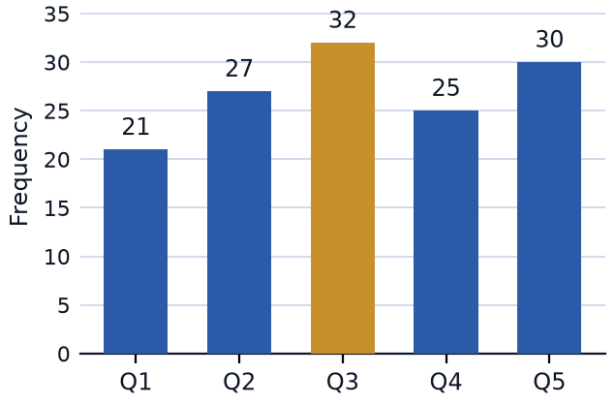
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15 [Challenging] • Target: Identifying the population and sample • Cross-Lesson: 3.04 Sampling Distributions & Central Limit Theorem

A hospital wants to estimate the average patient satisfaction score across its 12 different departments (e.g. cardiology, pediatrics, emergency), which are expected to have very different typical satisfaction levels. Describe, step by step, how you would design a stratified random sample to estimate overall patient satisfaction, and explain why a simple random sample of all patients might be less informative for comparing satisfaction across departments.

16 [Challenging] • Target: Identifying the population and sample • Cross-Lesson: 4.01 Confidence Intervals & Hypothesis Testing

A researcher wants to estimate the average time it takes customers to be served at a chain of 200 coffee shops nationwide. Due to budget constraints, the researcher can only visit a small number of physical locations. Describe how a cluster sample could be carried out in this situation, and explain a key risk of cluster sampling that would not apply if the researcher instead used a stratified sample based on region.



17 [Challenging] • Target: Wording and response bias • Cross-Lesson: 4.02 Chi-Square Tests & Regression Inference

A school district with 12 high schools wants to sample 300 teachers. Compare: (a) Stratified sampling (taking an SRS of 25 teachers from each of the 12 schools) vs. (b) Cluster sampling (randomly selecting 2 schools and surveying all 150 teachers in each). Discuss sampling variability and logistical practicality.

18 [Challenging] • Target: Comparing sample size and bias • Cross-Lesson: 1.01 Measures of Center & Spread

A town council is conducting a survey about a proposed new park. They decide to use a stratified random sample to ensure representation from different neighborhoods. The total population of the town is 45,000. The town is divided into three neighborhoods: Northside (18,000 residents), Eastside (15,000 residents), and Westside (12,000 residents). The council plans to survey a total of 900 residents. If the sample is to be proportional to the population of each neighborhood, how many residents should be surveyed from the Northside neighborhood?

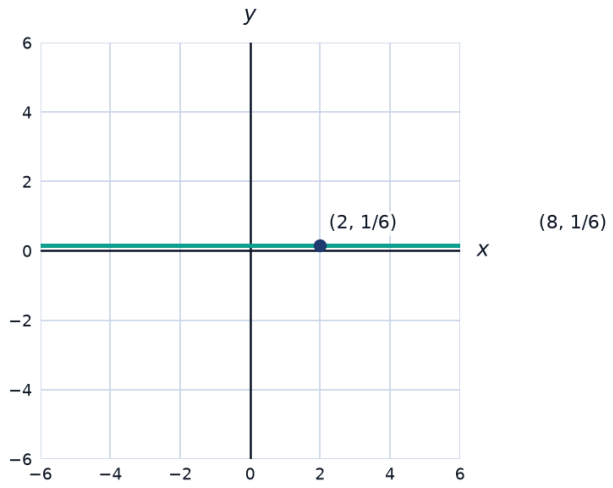
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19 [Challenging] • Target: Designing a sampling plan • Cross-Lesson: 1.02 Z-Scores & Linear Transformations

A diagnostic medical screening test is 95% sensitive (positive if disease present) and 90% specific (negative if disease absent). In a population, 2% of people have the disease. Find the probability that a person who tests positive actually has the disease.

20 [Challenging] • Target: Convenience sampling and its pitfalls • Cross-Lesson: 1.03 Describing Distributions & Linear Regression

Consider a continuous random variable X with a uniform distribution over the interval $[2, 8]$. What is the probability that X is between 4 and 7, i.e., $P(4 < X < 7)$?



Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Identifying sampling methods

A factory assigns each worker a number from 1 to 1223, then uses a random number generator to select 70 numbers, surveying those workers. Identify this sampling method.

a. Convenience sample

b. Simple random sample (SRS)

c. Stratified random sample

d. Cluster sample

22 [Foundational] • Target: Cluster sampling and systematic sampling

A researcher randomly selects 3 entire delivery routes out of 33 at a shipping company, and surveys every member within those selected delivery routes. Identify this sampling method.

a. Stratified random sample

b. Simple random sample (SRS)

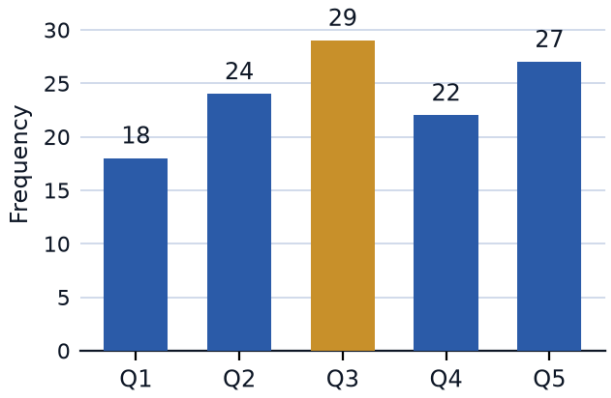
c. Systematic sample

d. Cluster sample

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23 [Foundational] • Target: Sources of bias

A researcher stands outside a farmers market and surveys the first 44 people encountered, intending to represent the general population. Identify this sampling method.



a. Systematic sample

b. Cluster sample

c. Simple random sample (SRS)

d. Convenience sample

24 [Foundational] • Target: Wording and response bias

A researcher divides a hospital into groups (department A, department B, and department C), then takes a simple random sample of 72 patients from each group. Identify this sampling method.

a. Cluster sample

b. Stratified random sample

c. Convenience sample

d. Simple random sample (SRS)

25 [Foundational] • Target: Comparing sample size and bias

A researcher stands outside a train station and surveys the first 49 people encountered, intending to represent the general population. Identify this sampling method.

a. Cluster sample

b. Simple random sample (SRS)

c. Convenience sample

d. Systematic sample

26 [Foundational] • Target: Designing a sampling plan

A researcher divides a factory into groups (freshmen, sophomores, juniors, and seniors), then takes a simple random sample of 67 workers from each group. Identify this sampling method.

a. Cluster sample

b. Systematic sample

c. Stratified random sample

d. Simple random sample (SRS)

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27 [Foundational] • Target: Convenience sampling and its pitfalls

A quality-control inspector at a clinic checks every 32nd item coming off a production line, starting from a randomly chosen point among the first 32 items. Identify this sampling method.

a. Simple random sample (SRS)

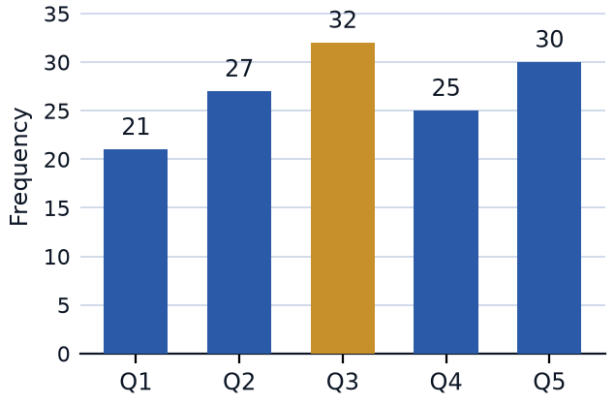
b. Convenience sample

c. Systematic sample

d. Cluster sample

28 [Foundational] • Target: Comparing sampling methods side by side

A survey about a new dress code is voluntarily filled out online by residents, and only 20% respond. Identify the type of bias most likely present.



a. Wording bias (leading question)

b. Convenience sampling bias

c. Voluntary response bias

d. Response bias (social desirability)

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 **[Intermediate]** • **Target: Identifying the population and sample**

A survey question reads: "Don't you agree that the wasteful dress code should be changed?" Identify the type of bias in this question.

a. Convenience sampling bias

b. Voluntary response bias

c. Wording bias (leading question)

d. Response bias (social desirability)

30 **[Intermediate]** • **Target: Probability Sampling Designs**

A survey about a new policy is voluntarily filled out online by employees, and only 14% respond. Identify the type of bias most likely present.

a. Convenience sampling bias

b. Undercoverage bias

c. Voluntary response bias

d. Response bias (social desirability)

31 **[Intermediate]** • **Target: Selection Bias & Undercoverage**

Respondents are asked directly, face-to-face, about tax evasion. Identify the type of bias likely to affect their answers.

a. Undercoverage bias

b. Response bias (social desirability)

c. Wording bias (leading question)

d. Convenience sampling bias

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32 **[Intermediate]** • **Target: Nonresponse and Response Bias**

A survey about a proposed schedule change is voluntarily filled out online by employees, and only 8% respond. Identify the type of bias most likely present.

a. Voluntary response bias

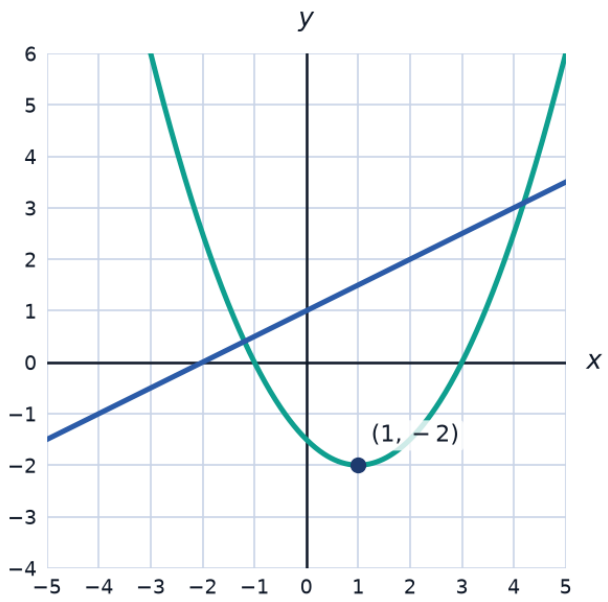
b. Response bias (social desirability)

c. Wording bias (leading question)

d. Convenience sampling bias

33 **[Intermediate]** • **Target: Parameters vs. Statistics**

Respondents are asked directly, face-to-face, about illegal drug use. Identify the type of bias likely to affect their answers.



a. Undercoverage bias

b. Convenience sampling bias

c. Response bias (social desirability)

d. Voluntary response bias

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34 [Intermediate] • Target: Identifying sampling methods

A researcher surveys people at a train station and generalizes the results to the entire city's population. Identify the type of bias most likely present.

a. Voluntary response bias

b. Wording bias (leading question)

c. Convenience sampling bias

d. Response bias (social desirability)

35 [Intermediate] • Target: Cluster sampling and systematic sampling

At a school with 35 homeroom classes, a researcher randomly selects 4 entire homeroom classes and surveys every individual within them. Identify this sampling method.

a. Systematic sample

b. Convenience sample

c. Cluster sample

d. Simple random sample (SRS)

36 [Intermediate] • Target: Sources of bias

At a hospital with 60 wards, a researcher randomly selects 3 entire wards and surveys every individual within them. Identify this sampling method.

a. Simple random sample (SRS)

b. Convenience sample

c. Cluster sample

d. Stratified random sample

37 [Intermediate] • Target: Wording and response bias

A clinic has 25805 enrolled patients in total. A researcher surveys a random sample of 873 patients to estimate average commute time. Find the size of the population in this scenario.

a. 25705

b. 25805

c. 25905

d. 873

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Comparing sample size and bias • Cross-Lesson: 1.01 Measures of Center & Spread

A company has 44627 enrolled employees in total. A researcher surveys a random sample of 422 employees to estimate average commute time. Find the size of the population in this scenario.

a. 44727

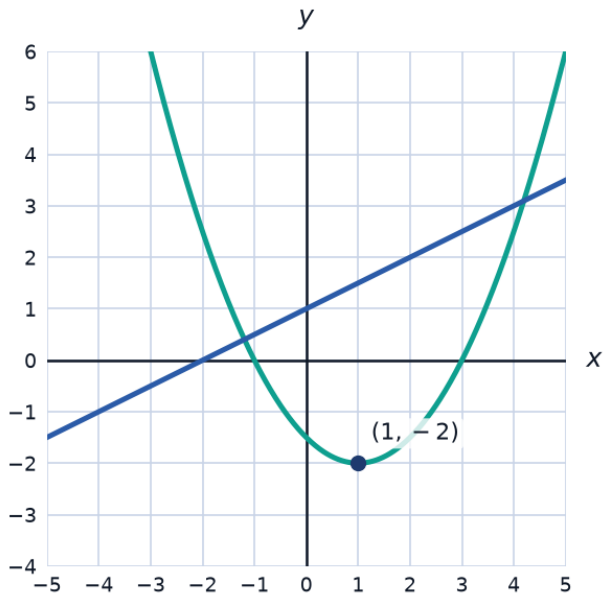
b. 44627

c. 422

d. 44527

39 [Challenging] • Target: Designing a sampling plan • Cross-Lesson: 1.02 Z-Scores & Linear Transformations

A pollster has a complete numbered list of all registered voters and wants every voter to have an exactly equal chance of selection. Recommend the most appropriate sampling method.



a. Systematic sample

b. Stratified random sample

c. Simple random sample (SRS)

d. Convenience sample

40 [Challenging] • Target: Convenience sampling and its pitfalls • Cross-Lesson: 1.03 Describing Distributions & Linear Regression

A factory manager wants a cheap, logistically convenient way to check product quality by inspecting a few whole production batches out of many. Recommend the most appropriate sampling method.

a. Systematic sample

b. Stratified random sample

c. Cluster sample

d. Simple random sample (SRS)

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41 [Challenging] • Target: Probability Sampling Designs • Cross-Lesson: 2.02 Experimental Design & Scope of Inference

Which sampling method gives every possible group of n individuals an equal chance of being chosen?

a. Simple Random Sample (SRS)

b. Stratified Random Sample

c. Cluster Sample

d. Systematic Sample

42 [Challenging] • Target: Probability Sampling Designs • Cross-Lesson: 3.01 Probability Rules & Conditional Probability

A city divides its territory into 40 voting precincts. Five precincts are randomly chosen, and every resident in those 5 precincts is surveyed. What sampling method is this?

a. Cluster sampling

b. Stratified sampling

c. Simple random sampling

d. Multistage convenience sampling

43 [Challenging] • Target: Selection Bias & Undercoverage • Cross-Lesson: 3.02 Discrete Random Variables & Binomial Distributions

Can increasing the sample size from 100 to $10,000$ eliminate bias in a voluntary response survey?

a. No, a larger biased sample simply estimates the biased quantity more precisely

b. Yes, large samples eliminate all forms of sampling bias by the Law of Large Numbers

c. Yes, provided $n \geq 1000$

d. Only if the population is normally distributed

44 [Challenging] • Target: Nonresponse and Response Bias • Cross-Lesson: 3.03 The Normal Distribution

A police officer in uniform asks high school students face-to-face if they have ever consumed alcohol. Which bias will most likely distort the findings?

a. Response bias, as students are likely to lie to authority figures

b. Selection bias

c. Voluntary response bias

d. Cluster bias

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45 [Challenging] • Target: Parameters vs. Statistics • Cross-Lesson: 3.04 Sampling Distributions & Central Limit Theorem

In statistical notation, which symbol represents a population parameter rather than a sample statistic?

a. μ

b. $\bar{x}$

c. s

d. $\hat{p}$

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